Lecture 9

Ribbon categories, Temperley–Lieb and $\mathrm{SU}(2)_k$

Objectives. *Lecture 7 found the topological spin as a closed curl, and Lecture 8 the matrix $c = A^{-1}B_q$ of the braiding on two spins. This lecture puts the two together. A twist makes a braided category draw ribbons, not strings, and then every framed knot has a value. On tensor powers of one spin the calculus is the Temperley–Lieb algebra, the value is the Kauffman bracket, and its projectors are the higher spins. At a root of unity one projector vanishes in every closed picture, and discarding it leaves a finite category: $\mathrm{SU}(2)_k$, the source of the Ising-like and Fibonacci-like anyons of Lecture 10.*

By the end of this lecture you will be able to:

- Define a twist and a ribbon category, and show that the curl of Lecture 7 satisfies the twist axiom.
- Compute with the Temperley–Lieb skein relation, and prove that the Kauffman bracket is invariant under Reidemeister II and changes by $-A^{\pm 3}$ under Reidemeister I.
- Construct the Jones–Wenzl projector p_2 and compute traces and twists of coloured strands.
- Explain how $\mathrm{SU}(2)_k$ arises at $A = e^{i\pi/(2(k+2))}$ by discarding negligible morphisms and flipping the pivotal structure.

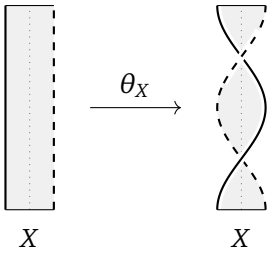
9.1 Twists and ribbon categories

Definition 9.1 (Twist, ribbon category)**.** A *twist* on a braided rigid category is a natural isomorphism $\theta_X : X \to X$ with $\theta_1 = \mathrm{id}$ and

$$\theta_{X\otimes Y} = c_{Y,X}\, c_{X,Y}\, (\theta_X \otimes \theta_Y).$$

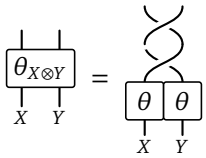
It is a *ribbon structure* if moreover $\theta_{X^*} = (\theta_X)^*$ (EGNO Def. 8.10.1; Kassel Def. XIV.3.2).

A ribbon carries a transverse direction as well as a centreline. The following band has one solid edge and one dashed edge, so that a full turn can be followed even when its centreline has not moved.



The untwisted band and the band with one full twist have the same dotted centreline. The marked edges distinguish their framings. The two edge crossings in the projection record one full turn of the band; they are not crossings of two independent object strands. The morphism θ_X records this turn, using the positive-twist convention of (9.2). Its mirror represents θ_X^{-1}.

The axiom is the picture of a ribbon, or a belt: a full turn of a belt holding two strands turns each strand and also wraps them once around each other,



(9.1)

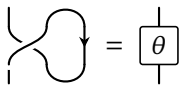
For a simple object θ_a is a number, and $\theta_{a\otimes b}$ acts on the channel c by θ_c; so the axiom is the ribbon relation $R_c^{ab}R_c^{ba} = \theta_c/(\theta_a\theta_b)$ that Lecture 7 cited.

Theorem 9.2 (The curl is a twist · *proved given coherence*)**.** *In a braided pivotal category, the curl of Lecture 7, the right partial trace of $c_{X,X}$, is a twist. In a fusion category with its canonical spherical structure, its value on a simple object is the topological spin θ_a of Theorem 7.7.*

Proof. Naturality: boxes slide through crossings, cups and caps. Invertibility: the curl drawn with the inverse crossing and the cap on the other side is its inverse, by RII and a snake. For $X \otimes Y$, draw the curl of the pair as a belt: the crossing $c_{X\otimes Y,X\otimes Y}$ is four crossings by the hexagons, and closing the right pair of strands with the nested cup and cap of Proposition 4.3 leaves a curl on each strand and the two crossings of $c_{Y,X}c_{X,Y}$ between them, which is the axiom. The value on a simple object is Theorem 7.7, which assumes the canonical structure. ∎

In another pivotal structure the curl can differ by a sign (§ 9.5). In a braided fusion category the curl is a ribbon structure exactly when the pivotal structure is spherical (EGNO Prop. 8.10.12); beyond fusion categories this can fail.

So framing matters. A closed curl is not a straight strand, as it would be in a plane drawing of a knot (Reidemeister I); with a twist, *framed* RI holds:



(9.2)

The strands of a ribbon category are thin bands lying flat in the page (blackboard framing), and a curl puts a full twist into the band.

9.2 Tangles and the Reshetikhin–Turaev functor

A *framed tangle* is a piece of a framed oriented link in a slab, with endpoints on the bottom and top; up to isotopy, tangles form a ribbon category under stacking and juxtaposition.

Theorem 9.3 (Shum; Reshetikhin–Turaev · *cited*)**.** *Framed oriented tangles form the free ribbon category on one object: for every object V of a strict ribbon category $\mathcal{C}$ there is a unique strict ribbon functor from tangles to $\mathcal{C}$ sending an upward strand to V (Kassel Thm. XIV.5.1).*

A framed link is a tangle from nothing to nothing, so its image is an endomorphism of 1, a number when $\mathrm{End}(1) = \mathbb{C}$, as in a fusion category: a link invariant for every object of such a ribbon category. The pictures of these lectures are honest because of this theorem.

9.3 Temperley–Lieb and the Kauffman bracket

Take $V = V_1$ of $U_q(\mathfrak{sl}_2)$, $q = A^2$, with the cup $v = e_1 \otimes e_2 - q\, e_2 \otimes e_1$ of Lecture 8. The cap $u\colon V \otimes V \to \mathbb{C}$ is given by

$$u(e_1 \otimes e_1) = u(e_2 \otimes e_2) = 0,$$
$$u(e_1 \otimes e_2) = -q^{-1}, \qquad u(e_2 \otimes e_1) = 1.$$

Both snakes are straight, as in Lecture 4. The contraction $u(v) = -q^{-1} - q$ gives the closed loop:

$$\bigcirc \;=\; \delta = -A^2 - A^{-2} = -[2]_q, \tag{9.3}$$

where $[n]_q = (q^n - q^{-n})/(q - q^{-1})$; at $A = \pm 1$ it is the Penrose loop -2 of Lecture 4. Write $U = v \circ u$, a cap followed by a cup. Then $U^2 = \delta U$, and on three strands $U_1U_2U_1 = U_1$, $U_2U_1U_2 = U_2$: the *Temperley–Lieb* relations. The *Temperley–Lieb algebra* $\mathrm{TL}_n(\delta)$ has generators $U_1, \dots, U_{n-1}$, these relations, and $U_iU_j = U_jU_i$ for $|i - j| \geq 2$; by Kauffman's theorem it is the algebra of the diagrams $n \to n$ below (Kauffman–Lins p. 9).

Definition 9.4 (The Temperley–Lieb category)**.** $\mathrm{TL}(\delta)$ has objects $n = 0, 1, 2, \dots$ and as morphisms $n \to m$ the linear combinations of planar diagrams of non-crossing strands joining n points below to m above; composition stacks diagrams and replaces each closed loop by δ.

Theorem 9.5 (Temperley–Lieb acts on spin $\frac{1}{2}$)**.** *With $\delta = -[2]_q$, $U_i \mapsto \mathrm{id} \otimes U \otimes \mathrm{id}$ (acting on strands $i, i+1$) defines a representation of $\mathrm{TL}_n(\delta)$ on $V_1^{\otimes n}$ by U_q-module maps.*

Proof. $U = v \circ u = B_q - q$ (multiply out) is a module map because $B_q = Ac$ is (Lecture 8). Far-apart generators act on different factors and commute. The relation $U^2 = \delta U$ follows from $u \circ v = \delta$. For the adjacent relation, expand each U as $v \circ u$. The two middle composites are the following snakes:



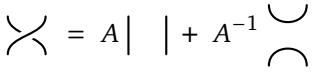
$S_L = (u \otimes \mathrm{id})(\mathrm{id} \otimes v)$ $\qquad$ $S_R = (\mathrm{id} \otimes u)(v \otimes \mathrm{id})$

Both S_L and S_R equal id_V, by the straight-snake identities for u and v. Associativity of composition and interchange give

$$\begin{aligned} U_1U_2U_1 &= (v\otimes \mathrm{id})\, S_L\, S_R\, (u\otimes \mathrm{id}) \\ &= (v\otimes \mathrm{id})(u\otimes \mathrm{id}) = U_1. \end{aligned}$$

Reflecting this calculation left to right gives $U_2U_1U_2 = U_2$. Thus all the defining TL relations hold locally on the tensor factors. ∎

The braiding is then a combination of planar diagrams: from $c = A^{-1}B_q$ and $B_q = q + U$, the *skein relation*



(9.4)

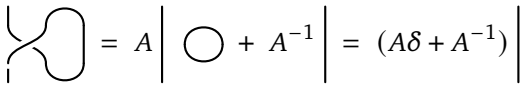
and the inverse crossing exchanges A and A^{-1}. This is the Kauffman bracket's rule as in Simon (Fig. 2.3) and Kauffman–Lins (p. 5); many references, and mirror images, exchange A and A^{-1}.

Theorem 9.6 (The Kauffman bracket · *sketched; proved for RI and RII*)**.** *Evaluating a link diagram by* (9.4) *and* (9.3)*, the value $\langle L\rangle$ is unchanged by Reidemeister II and III, and a curl multiplies it by $-A^3$, or $-A^{-3}$ for the opposite curl.*

Proof. RII: a crossing followed by its inverse is

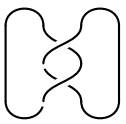
$$\begin{aligned} &(A + A^{-1}U)(A^{-1} + AU) \\ &= 1 + (A^2 + A^{-2})U + U^2 \\ &= 1 + (A^2 + A^{-2} + \delta)U = 1. \end{aligned}$$

RI: expanding the crossing of a curl gives



and $A\delta + A^{-1} = -A^3$. RIII is the braid relation of Theorem 7.4 for $c = A + A^{-1}U$, which follows from the TL relations alone (Sheet 9). ∎

So $\langle L\rangle$ is an invariant of framed links, and the factor $-A^3$ is the twist θ_{V_1}. For an oriented link, the *writhe* $w(L)$ is the sum of crossing signs. Our /-over crossing of two upward strands has sign +1; reversing one strand reverses the sign. Let O denote the unknot. Then $f_L(A) = (-A^3)^{-w(L)}\langle L\rangle/\langle O\rangle$ is an invariant of unframed links, and at $A = t^{-1/4}$ it is the *Jones polynomial* $V_L(t)$ (Kauffman–Lins Thm. 1). For the Hopf link with two positive crossings,



(9.5)

Indeed, squaring $c = A + A^{-1}U$ and using $U^2 = \delta U$ gives

$$c^2 = A^2 + (2 + A^{-2}\delta)U.$$

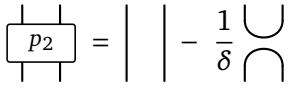
Closing the identity gives two loops. Closing U gives one. Hence

$$\begin{aligned} \langle L\rangle &= A^2\delta^2 + (2 + A^{-2}\delta)\delta \\ &= \delta(-A^4 - A^{-4}). \end{aligned}$$

Thus $\langle L\rangle/\langle O\rangle = -A^4 - A^{-4}$. Sheet 9 does the trefoil.

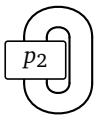
9.4 p_2 and coloured strands

Spin 1 sits inside $V_1 \otimes V_1$ as the image of a projection that kills the cup:



(9.6)

Then $p_2^2 = p_2$ and $Up_2 = 0 = p_2U$, whenever $\delta \neq 0$. Its trace closes both strands:



$$= \delta^2 - \frac{\delta}{\delta} = \delta^2 - 1 = [3]_q, \tag{9.7}$$

the dimension of spin 1, deformed; at $q = 1$, p_2 is the symmetriser $\frac{1}{2}(1 + P)$ of the triplet, the symmetric part of $\mathbb{C}^2 \otimes \mathbb{C}^2$ (*Symmetries* Ex. 14.13).

The image of p_2 is an ordinary subspace in the representation model. For example, at $q = 1$ the matrix acts by

$$p_2 \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} x_1 \\ (x_2 + x_3)/2 \\ (x_2 + x_3)/2 \\ x_4 \end{pmatrix}.$$

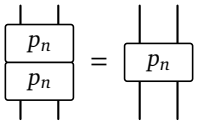
Its image is the triplet, spanned by $e_1 \otimes e_1$, $e_1 \otimes e_2 + e_2 \otimes e_1$ and $e_2 \otimes e_2$. Applying p_2 again fixes every vector in this image. Thus the path from projector to object is: the endomorphism p_2 of two strands selects its image, and the coloured strand $(2, p_2)$ names that image as an object in its own right. The original TL category has only the objects n; its idempotent completion adds a name for each such selected part.

Theorem 9.7 (Jones–Wenzl projectors · *cited; proved for* p_2)**.** *On $n+1$ strands, starting from $p_1 = \mathrm{id}$,*

$$p_{n+1} = p_n \otimes 1 + \frac{[n]_q}{[n+1]_q} (p_n \otimes 1)\, U_n\, (p_n \otimes 1)$$

defines idempotents killed by every U_i, for as long as $[m]_q \neq 0$ for $m \leq n+1$ (Kauffman–Lins Def. 1 and Lemma 1; for p_2 it is (9.6)). Their traces are $\operatorname{tr} p_n = (-1)^n [n+1]_q$ (ibid., p. 13).

In pictures, a projector absorbs a second copy of itself,



(9.8)

and is killed by any cap across two of its strands:



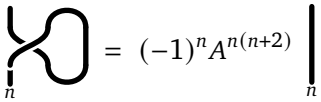
(9.9)

A pair (n, p_n), n strands carrying the idempotent p_n, is a *coloured strand*, drawn as one strand labelled n. It is an object of the *idempotent completion*: objects (X, p) with $p^2 = p$, morphisms $f\colon X \to Y$ with $f = p_Y f p_X$. Adding formal direct sums of such pairs gives

the *additive* idempotent completion. This is how spin $n/2$ enters the diagram calculus. A coloured strand has a twist, from the full twist of its n strands,

$$\theta_n = (-1)^n A^{n(n+2)}$$

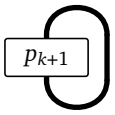
(Kauffman–Lins Prop. 6), and $\theta_1 = -A^3$. In pictures, a curl on a coloured strand is that number:



(9.10)

9.5 Roots of unity, the negligible quotient, the pivotal flip

Fix $k \geq 1$ and $A = e^{i\pi/(2(k+2))}$, so $q = A^2 = e^{i\pi/(k+2)}$. Then $[k+2]_q = 0$ while $[m]_q \neq 0$ for $m \leq k+1$: the recursion reaches p_{k+1}, and $\operatorname{tr} p_{k+1} = (-1)^{k+1}[k+2]_q = 0$:



(9.11)

Definition 9.8 (Negligible). A morphism $f\colon X \to Y$ is *negligible* if $\operatorname{tr}(gf) = 0$ for every $g\colon Y \to X$.

The negligible morphisms form an ideal: closed under composition and $\otimes$ with anything (EGNO Exer. 8.18.9(ii)). By Kauffman–Lins (Lemma 5), p_{k+1} closes to 0 in every network: it is negligible.

A zero trace is only one closure test. For instance, in $\mathrm{Vect}_{\mathrm{fd}}$ the map $f = \operatorname{diag}(1,-1)$ has trace zero, but $g = \operatorname{diag}(1,0)$ gives $\operatorname{tr}(gf) = 1$. So this f is not negligible. The cited result about p_{k+1} tests every closure, not just $\operatorname{tr}(p_{k+1})$.

The quotient at $k = 1$. Here $A = e^{i\pi/6}$, $q = e^{i\pi/3}$ and $\delta = -1$. The projector is $p_2 = 1 + U$, a nonzero diagrammatic endomorphism before the quotient. Every endomorphism of two strands is $a\,1 + bU$. The relations $Up_2 = 0$ and $\operatorname{tr}(p_2) = \delta^2 - 1 = 0$ give

$$\operatorname{tr}\big((a\,1 + bU)p_2\big) = 0 \qquad (a, b \in \mathbb{C}).$$

Thus p_2 is negligible in this case by the definition itself. In the idempotent completion it is the identity of the object $(2, p_2)$. Setting that identity to zero makes this object a zero object in the quotient. It does not assert that the matrix of p_2 was zero before quotienting.

The complementary projector $1 - p_2 = -U$ remains. The cup $v\colon 0 \to 2$ and the cap $-u\colon 2 \to 0$ identify this surviving part with the tensor unit, because in the quotient

$$(-u)v = 1_0, \qquad v(-u) = -U = 1_2.$$

Write s for the surviving one-strand colour 1. There are two simple objects, the unit and s, with $s \otimes s \cong 1$, as the general theorem below confirms. The quotient removes a channel. A subsequent pivotal change alters its closing and twist conventions: the raw values $d_s = -1$ and $\theta_s = -A^3 = -i$ become $d_s = 1$ and $\theta_s = i$ after multiplication by $(-1)^1$. These are different operations.

Theorem 9.9 (SU(2)$_k$ · *cited*)**.** *The additive idempotent completion of* TL(δ) *at* $A = e^{i\pi/(2(k+2))}$*, modulo negligible morphisms, is a semisimple ribbon category with simple objects the coloured strands* $0, 1, \dots, k$*, fusing by the truncated Clebsch–Gordan rule of Lecture 5 (Kauffman–Lins Lemmas 4–5; Simon Ch. 22, Tables 22.1–22.3). As a tensor category it is EGNO's* $C_k(q)$ *(Ex. 8.18.5): the quotient of the tilting* U_q*-modules by the negligible ones (Bakalov–Kirillov Thm. 3.3.20). Here "tilting modules" names that cited alternative construction; our construction uses the TL diagrams and projectors above. In the pivotal structure of* TL*,*

$$d_n = (-1)^n [n+1]_q,$$
$$\theta_n = (-1)^n A^{n(n+2)}.$$

These d_n are negative for odd n: the Penrose sign of Lecture 4, now on every half-integer spin. The cure is the one Lecture 4 found, a different pivotal structure. Multiplying the pivotal structure by $(-1)^n$ on the colour n is allowed, since $(-1)^n$ is a monoidal function of n (the truncated rule preserves parity); it multiplies d_n and θ_n by $(-1)^n$ (Bakalov–Kirillov Ex. 2.2.8; Simon (15.1)). With $j = n/2$:

$$d_j = [2j+1]_q > 0,$$
$$[2j+1]_q = \frac{\sin\big((2j+1)\pi/(k+2)\big)}{\sin\big(\pi/(k+2)\big)},$$
$$\theta_j = e^{2\pi i\, j(j+1)/(k+2)}.$$

At $k = 2$ the simple objects are 0, $\frac{1}{2}$ and 1, the coloured strands $0, 1, 2$; the truncated rule gives $\frac{1}{2} \otimes \frac{1}{2} \cong 0 \oplus 1$, $\frac{1}{2} \otimes 1 \cong \frac{1}{2}$ (since p_3, which would carry spin $\frac{3}{2}$, is negligible) and $1 \otimes 1 \cong 0$. This is the unitary SU(2)$_k$ of Lecture 5, with the dimensions found there as Frobenius–Perron dimensions (Bonderson §5.4). At $k = 1$, $\theta_{1/2} = -A^3 = -i$ in the TL structure and $+i$ after the flip: the semion of Lecture 7. At $k = 2$, $d_{1/2} = \sqrt{2}$ and $\theta_{1/2} = e^{3\pi i/8}$, Kitaev's $\nu = 3$: the $\kappa = -1$ category of Lecture 6, not the course's Ising. At $k = 3$, the integer spins 0 and 1 close under fusion, $1 \otimes 1 = 0 \oplus 1$, with $d_1 = [3]_q = \varphi$ and $\theta_1 = e^{4\pi i/5}$: Fibonacci, with the chirality chosen in Lecture 7. The other choice $-A$ of the root multiplies θ_j by $(-1)^{2j}$, and reverses the sign of the braiding $c = A^{-1}B_q$ on two spins.

Why is the quotient semisimple? Each coloured strand $n \le k$ has non-zero trace, so its identity is not negligible and it stays a simple object; the morphisms that would make the category non-semisimple, through p_{k+1} and beyond, are negligible and are set to zero. The traces that survive are the dimensions above.

Summary.

- *Handed on:* a twist satisfies $\theta_{X\otimes Y} = c_{Y,X}c_{X,Y}(\theta_X \otimes \theta_Y)$, and the curl is one (Theorem 9.2); framed tangles form the free ribbon category (Theorem 9.3).
- *Temperley–Lieb:* on $V_1^{\otimes n}$ the crossing is $A \cdot \mathrm{id} + A^{-1} \cdot (\mathrm{cup} \circ \mathrm{cap})$, the loop is $-A^2 - A^{-2}$, and the Kauffman bracket is a framed link invariant (Theorem 9.6).
- *Projectors:* for $\delta \neq 0$, $p_2 = 1 - U/\delta$. The Jones–Wenzl recursion reaches p_n while $[m]_q \neq 0$ for $1 \le m \le n$, with $\mathrm{tr}\, p_n = (-1)^n[n+1]_q$ and $\theta_n = (-1)^n A^{n(n+2)}$ (Theorem 9.7).
- SU(2)$_k$*:* at $A = e^{i\pi/(2(k+2))}$, p_{k+1} is negligible; the quotient, with the pivotal flip, has $d_j = [2j+1]_q$ and $\theta_j = e^{2\pi i j(j+1)/(k+2)}$ (Theorem 9.9).
- *Next:* Lecture 10 adds the S-matrix and makes these categories modular.

References

Etingof, Gelaki, Nikshych and Ostrik, *Tensor Categories*: twists and ribbon structures, Def. 8.10.1 and Rem. 8.10.3, p. 216; ribbon iff spherical, Prop. 8.10.12, p. 220; the pivotal flip, Rem. 8.10.16, p. 221; the categories $C_k(q)$ and negligible morphisms, Ex. 8.18.5 and Exer. 8.18.9, pp. 235–236 (their twist $q^{-m(m+2)/2}$ is the inverse of ours). Kassel, *Quantum Groups*: framed tangles, §X.8, pp. 270–272; ribbon categories, Def. XIV.3.2, p. 349; the universal ribbon category, Thm. XIV.5.1, p. 360. Bakalov and Kirillov, *Lectures on Tensor Categories and Modular Functors*: ribbon categories, Def. 2.2.1 and Ex. 2.2.8, pp. 33–35; Reshetikhin–Turaev, Thm. 2.3.10, p. 42; $\mathrm{SU}(2)_k$, Thm. 3.3.20, p. 67. Kauffman and Lins, *Temperley–Lieb Recoupling Theory*: the bracket, pp. 5–9 and Thm. 1; the Jones–Wenzl recursion, Def. 1 and Lemma 1, pp. 13–14; the root of unity, Lemmas 4–5, pp. 37–38; twisted projectors, Prop. 6, p. 43. Simon, *Topological Quantum*: the skein relation, Fig. 2.3, p. 4; twists, Ch. 15, pp. 199–200; Temperley–Lieb–Jones, Ch. 22, pp. 297–310, and the theories at $A = \pm e^{i\pi/(2(k+2))}$, Tables 22.1–22.3. Bonderson, PhD thesis, §5.4, p. 81.